6 The pick-up on an electric guitar produces an electrical signal from the vibrations of the guitar string. The pick-up consists of a small coil of insulated wire wound round a small cylindrical magnet as shown in Fig. 6.1.

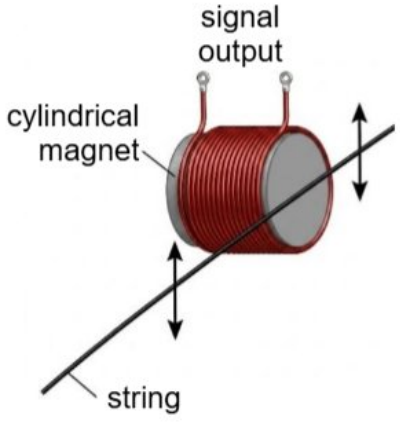


Fig. 6.1

The strings of the guitar are made of steel, which is a conductor. When a string vibrates, an electrical signal is generated between the terminals of the coil.

(a) Use Faraday's Law to explain why an electrical signal is generated.

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.....[3]

(b) Suggest why

(i) the electrical signal needs to be amplified before connection to a loudspeaker,

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.....[1]

(ii) the design of the pick-up would be inappropriate for use on a guitar with nylon strings.

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.....[2]

[Total: 6]